



Subnet Segmentation

Manufacturing ERP System

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Client: Manufacturer

Estimated Duration: ~3 Weeks

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1. Executive Summary

This document outlines the network segmentation for the Manufacturing ERP System using Zero Trust principles. The design ensures workload isolation, least-privilege access, defense in depth, and scalability.

1) Network Address Plan

VNet	CIDR	Subnet	Purpose
Hub VNet	10.0.0.0/16	GatewaySubnet /27	VPN Gateway
		AzureFirewallSubnet /26	Azure Firewall
Spoke-DB	10.1.0.0/16	SQL Database /24	SQL VMs
Spoke-App	10.2.0.0/16	Web Application /24	IIS Web Servers
Spoke-Storage	10.3.0.0/16	Private Endpoints /24	Storage endpoints
DR Hub	10.10.0.0/16	AzureFirewallSubnet /26	DR Firewall

3. Primary Region Subnets

3.1 Hub VNet Subnets (10.0.0.0/16)

Subnet	CIDR	Usable Range	Purpose	NSG/Route Table
GatewaySubnet	10.0.1.0/27	10.0.1.4-10.0.1.30	VPN Gateway	None
AzureFirewallSubnet	10.0.2.0/26	10.0.2.4-10.0.2.62	Azure Firewall	None

3.2 Database Spoke Subnets (10.1.0.0/16)

Subnet	CIDR	Usable Range	Purpose	NSG/Route Table
SQL Database Subnet	10.1.1.0/24	10.1.1.4-10.1.1.254	SQL Server VMs	NSG: nsg-sql

3.3 Application Spoke Subnets (10.2.0.0/16)

Subnet	CIDR	Usable Range	Purpose	NSG/Route Table
Web Application Subnet	10.2.1.0/24	10.2.1.4-10.2.1.254	IIS Web Servers	NSG: nsg-web

3.4 Storage Spoke Subnets (10.3.0.0/16)

Subnet	CIDR	Usable Range	Purpose	NSG/Route Table
Private Endpoints Subnet	10.3.1.0/24	10.3.1.4-10.3.1.254	Storage Endpoints	NSG: nsg-pe

4. DR Region Subnets

4.1 DR Hub Subnets (10.10.0.0/16)

Subnet	CIDR	Purpose
GatewaySubnet	10.10.1.0/27	VPN Gateway
AzureFirewallSubnet	10.10.2.0/26	DR Firewall

4.2 ASR-Managed Subnets

Original Subnet	DR Subnet	DR CIDR
snet-sql (10.1.1.0/24)	snet-sql-asr	10.1.1.0/24
snet-web (10.2.1.0/24)	snet-web-asr	10.2.1.0/24

5. NSG Best Practices Applied

Priority Range	Purpose
100-199	Application-specific allow rules
200-299	Azure service allow rules
300-399	Management access rules
400-499	Security/logging rules
4096	Default deny all